



8 – ELEMENT QUAD BAND ANTI-JAMMING ANTENNA

MAJOR PROJECT-A REPORT

**Submitted in Partial Fulfilment of the Requirements of the Degree of Bachelor
of Technology in Electronics and Telecommunication Engineering**

By

HIMANI MUKUND BAMNIKAR (TY/A1-06)

SAMRUDDHI TUKARAM CHAVAN (TY/A1-11)

ANSH VISHWAS TARALEKAR (TY/A3-59)

**Under the Guidance of
DR. KIRAN RATHOD**



Department of Electronics and Telecommunication Engineering
K. J. Somaiya Institute of Technology, Ayurvihar, Sion (E.), Mumbai 400022.
An Autonomous Institute permanently affiliated to University of Mumbai
University of Mumbai (2025-26)



CERTIFICATE

This is to certify that the project entitled “8- Element Quad Band Anti Jamming Antenna” is a Bonafide work of students Himani Mukund Banmikar, Samruddhi Tukaram Chavam, and Ansh Vishwas Taralekar submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of Bachelor of Technology in Electronics and Telecommunication Engineering.

Dr. Kiran Rathod (Project Guide)

Department of Electronics and Telecommunication Engineering

Dr. Jayashree Khanapuri
(Head of the Department)

Dr. Vivek Sunnapwar
(Principal)

Place: Sion, Mumbai-400022

Date: 5th May 2026

Seal of the College

MAJOR PROJECT-A APPROVAL

For

**Bachelor of Technology in Electronics and
Telecommunication Engineering**

This Project report entitled “*8- Element Quad Band Anti Jamming Antenna*”

By

Himani Mukund Bamnikar (TY/A1-06)

Samruddhi Tukaram Chavan (TY/A1-11)

Ansh Vishwas Taralekar (TY/A3-59)

is approved for the degree of Bachelor of Technology in Electronics and
Telecommunication Engineering.

Examiners:

1. _____

2. _____

3. _____

4. _____

Date: 5th May 2026

Place: Sion, Mumbai-400022

DECLARATION

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Himani Mukund Bamnikar _____

Samruddhi Tukaram Chavan _____

Ansh Vishwas Taralekar _____

Date: 5th May 2026

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Himani Mukund Bamnikar (TY/A1-06)

Samruddhi Tukaram Chavan (TY/A1-11)

Ansh Vishwas Taralekar (TY/A3-59)

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CHAPTER 1: INTRODUCTION

Navigation systems are very important in today's world and are widely used in communication, transportation, defence, aviation, and autonomous vehicles. Navigation with Indian Constellation (NavIC), developed by ISRO, is India's regional satellite navigation system that provides accurate Position, Navigation, and Timing (PNT) services. It is used in both civilian and military applications such as vehicle tracking, disaster management, and secure operations. With the increasing use of satellite-based navigation, ensuring reliable and accurate signal reception has become very important.

However, satellite signals become very weak when they reach the Earth due to long transmission distances. Because of this, they are highly affected by noise, interference, and intentional jamming. Jamming can block or disturb the original signals, leading to poor navigation performance. Conventional single-element antennas are not able to handle such interference effectively, which may result in loss of signal and reduced accuracy in critical applications.

To overcome these challenges, advanced antenna systems using multiple elements are used. These systems improve signal quality by receiving signals from different directions and processing them efficiently. Techniques like beamforming help to strengthen the desired signal, while null steering helps to suppress unwanted interference. This improves overall signal clarity and reliability.

In this project, an 8-element quad-band anti-jamming antenna is designed to operate on multiple frequency bands such as L1, L2, L5, and S-band. The system aims to improve signal strength, reduce interference, and provide reliable navigation even in difficult environments. It can be used in applications like defence, aviation, marine systems, and autonomous vehicles where accurate positioning is essential.

CHAPTER 2: OBJECTIVE

The primary objective of this project is to design, develop, and implement an efficient anti-jamming antenna system for NavIC-based navigation. The aim is to enhance the reliability and accuracy of satellite signals by reducing the effects of interference and jamming. Conventional single-element antennas are not capable of effectively handling interference; therefore, this project focuses on developing a multi-element antenna array with advanced signal processing techniques to ensure robust performance in challenging environments.

Another important objective is to design a compact quad-band antenna element that can operate efficiently across multiple frequency bands, namely L1, L2, L5, and S-band. These frequency bands are essential for NavIC communication, and supporting multiple bands increases system reliability. By combining multiple antenna elements into an 8-element array, the system aims to achieve spatial filtering, which helps in distinguishing between desired signals and interfering sources.

The project also aims to implement Direction of Arrival (DoA) estimation techniques to identify the direction from which interference signals are coming. Based on this information, advanced signal processing methods such as adaptive beamforming will be applied to enhance the desired NavIC signals. Additionally, null steering techniques will be used to suppress or eliminate unwanted signals, thereby improving overall signal quality and system performance.

Furthermore, the objective includes achieving good antenna performance parameters such as low return loss (S_{11} below -10 dB), Voltage Standing Wave Ratio (VSWR less than 2), and sufficient gain. The system will be designed and analysed using simulation tools, followed by hardware implementation using appropriate RF components. Finally, the project aims to validate the performance of the proposed system through testing using instruments like a Vector Network Analyzer (VNA) and spectrum analyser to ensure its effectiveness in real-world conditions.

CHAPTER 3: LITERATURE SURVEY

A comprehensive review of existing research has been carried out to understand current advancements and identify gaps in the field of anti-jamming antenna systems for GNSS/NavIC applications.

In [1], S. Li et al. proposed a robust phase compensation technique for GNSS antenna array receivers to mitigate phase distortions introduced by adaptive beamforming. The study emphasized maintaining carrier phase accuracy while achieving effective interference suppression. However, the work is primarily limited to simulation-based validation and does not fully address real-time implementation challenges.

In [2], Y. Xie et al. introduced a deep neural network-based anti-jamming approach for GNSS array receivers. Their method demonstrated improved performance in complex and dynamic interference environments compared to conventional adaptive beamforming techniques. Despite its effectiveness, the approach requires high computational resources and extensive training datasets, making real-time deployment challenging.

In [3], Y. Sun et al. proposed an anti-jamming method based on the physical rotation of antenna arrays. This technique enhances interference suppression by modulating spatial characteristics of received signals. However, the method introduces mechanical complexity and may not be suitable for fast-changing or real-time interference scenarios.

In [4], J. Arribas et al. investigated multi-antenna techniques for GNSS signal acquisition. The study demonstrated that antenna arrays improve robustness against interference and enhance signal acquisition performance. However, the work mainly focuses on acquisition and does not extensively address phase distortions or real-world dynamic jamming conditions.

CHAPTER 4: SOCIAL/MEDICAL/AGRICULTURAL/INDUSTRY RELEVANCE

The proposed anti-jamming antenna system is important across multiple sectors as it ensures reliable and uninterrupted navigation even in the presence of interference and jamming. With increasing dependence on satellite-based positioning, such robust and secure systems are essential for improving safety, efficiency, and continuous operation.

From a social perspective, the system plays an important role in disaster management and public safety by providing reliable navigation during emergencies. During natural disasters like floods, earthquakes, or cyclones, communication systems may fail, making satellite navigation essential for rescue operations. The proposed system ensures uninterrupted positioning, helping rescue teams quickly locate affected areas and reach victims efficiently, thereby improving overall response time and safety.

In the medical field, reliable navigation is important for tracking ambulances and ensuring they reach hospitals quickly. It also supports real-time tracking of medical supplies, blood, and organ transport where timing is critical. By providing accurate and continuous positioning, the system improves coordination between emergency services and hospitals, helping enhance healthcare efficiency and save lives.

In the agricultural sector, modern farming relies on precision agriculture using GPS/NavIC for tasks like ploughing, seeding, irrigation, and harvesting. Drones are also used for crop monitoring and spraying. These systems require accurate positioning, and the proposed anti-jamming system ensures reliable operation even in the presence of interference, improving crop management, productivity, and efficient use of resources.

From an industrial perspective, the system is useful in aviation, maritime navigation, defence, and UAV applications. It ensures accurate positioning for safe aircraft landing and ship navigation, while also protecting systems from jamming in defence operations. Additionally, it enables reliable navigation for drones used in surveillance, delivery, and mapping, ensuring safe and efficient operation.

Overall, the proposed anti-jamming antenna system enhances the reliability, safety, and efficiency of navigation systems across various sectors. It supports critical operations, reduces risks caused by signal interference, and contributes to technological advancement in modern society.

CHAPTER 5: BLOCK DIAGRAM AND ITS EXPLANATION

The proposed architecture features an 8-element quad-band antenna array (circular or planar) designed to capture NavIC satellite signals alongside potential interference. These signals are initially processed by an RF front-end—utilizing low-noise amplifiers, bandpass filters, and power dividers—to amplify weak data and eliminate noise.

Following analog-to-digital conversion, the system operates in the digital domain to mitigate disruptions:

- DoA Estimation: Determines the precise direction of interfering sources.
- Adaptive Beamforming: Combines antenna signals to focus on desired satellites.
- Null Steering & STAP: Uses Space-Time Adaptive Processing to dynamically adjust antenna weights, neutralizing interference while enhancing signal integrity.

System Architecture:

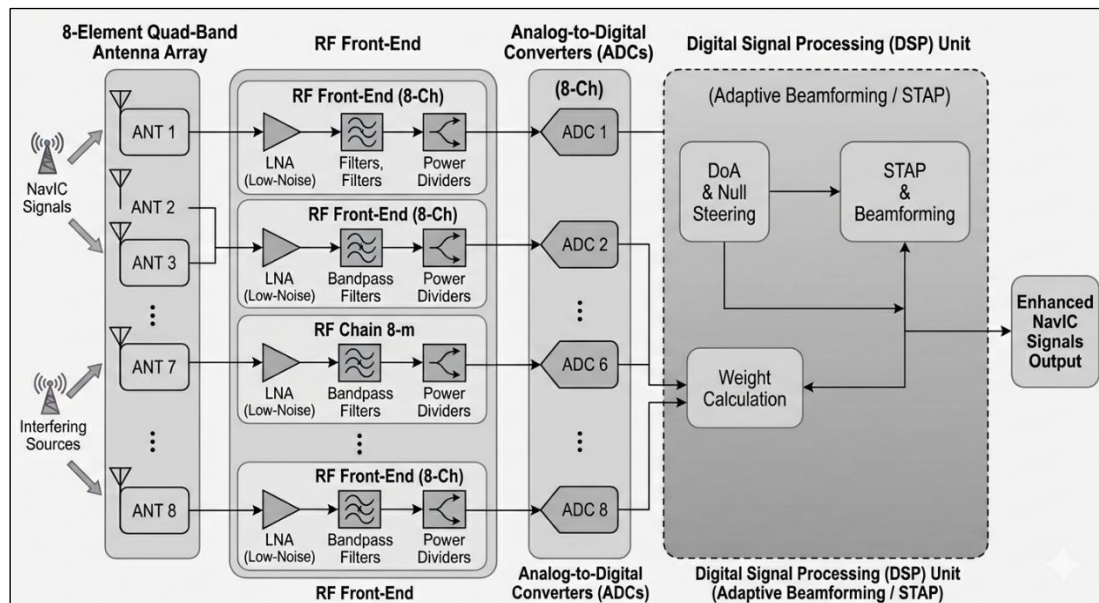


Fig 5.1: System architecture for Anti-Jamming NavIC Receiver.

The architecture of this anti-jamming system is built to ensure high-precision navigation by isolating legitimate satellite data from noise and intentional disruption. Here is a detailed, original breakdown of each functional stage:

1. 8-Element Quad-Band Antenna Array

- The system starts with 8 antenna elements (ANT1 to ANT8) arranged in an array.
- These antennas receive both NavIC signals (desired signals) and interfering/jamming signals from different directions.
- Using multiple antennas provides spatial diversity, which helps in identifying the direction of signals.
- Quad-band operation allows the system to work on multiple frequency bands (L1, L2, L5, S-band), improving reliability.

2. RF Front-End (Radio Frequency Stage)

Each antenna is connected to its own RF chain:

- LNA (Low Noise Amplifier):
 - Amplifies very weak incoming satellite signals.
 - Adds minimal noise to maintain signal quality.
- Bandpass Filter (BPF):
 - Allows only required frequency bands to pass.
 - Removes unwanted signals and noise from other frequencies.
- Power Divider:
 - Distributes the signal properly for further processing.
 - Ensures equal signal flow in each channel.

3. Analog-to-Digital Converters (ADCs)

- Each RF channel is connected to an ADC (ADC1 to ADC8).
- Converts analog signals into digital signals.
- This step is important because digital signals can be processed using algorithms.

4. Digital Signal Processing (DSP) Unit

This is the core part of the system where interference is removed.

a) Direction of Arrival (DoA) & Null Steering

- Determines the direction of incoming signals (both desired and interference).
- Helps identify where the jammer is located.
- Null steering creates nulls (zero response) in the direction of interference.

b) Weight Calculation

- Calculates weights (adjustment factors) for each antenna signal.
- These weights control how much each antenna contributes to the final signal.

c) STAP & Adaptive Beamforming

- Beamforming: Focuses on desired satellite signals.
- STAP (Space-Time Adaptive Processing):
 - Uses both spatial (antenna position) and time information.
 - Dynamically adjusts weights to suppress interference.
- Combines signals in such a way that:
 - Desired signals are strengthened
 - Interference is cancelled

5. Enhanced NavIC Signal Output

- Final output is a clean and strong signal.
- Interference and jamming are minimized.
- Provides accurate and reliable navigation data.

CHAPTER 6: SOFTWARE AND HARDWARE REQUIREMENTS

Software Requirements

The software implementation of the proposed anti-jamming antenna system involves the use of advanced simulation tools for accurate design, analysis, and optimization. HFSS (High Frequency Structure Simulator) and CST Microwave Studio are used for electromagnetic simulation and 3D modeling of the quad-band antenna and its array configuration. These tools help in evaluating important parameters such as return loss, gain, radiation pattern, bandwidth, and impedance matching, ensuring efficient and reliable antenna performance.

For signal processing, MATLAB is used as the main platform to implement algorithms such as adaptive beamforming, null steering, and Direction of Arrival (DoA) estimation. In addition, Python may be used for data analysis, visualization, and algorithm prototyping. The combination of these tools makes the development process more flexible, efficient, and suitable for accurate system design and testing.

Implementation Points:

- **Simulation & Modelling:** HFSS and CST are used to design and optimize antenna structure and array configuration.
- **Algorithm Development:** MATLAB is used for beamforming and DoA estimation, while Python is used for prototyping and data analysis.

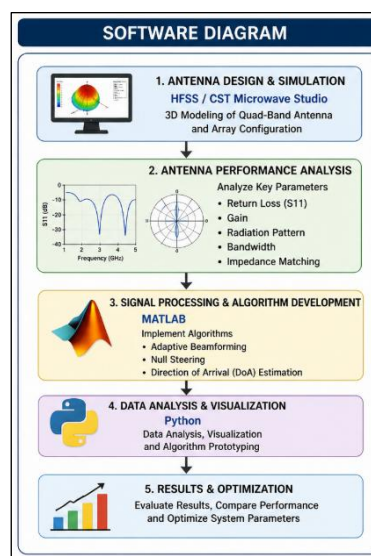


Fig 6.1: Software requirements.

Hardware Requirements

The hardware implementation requires a Printed Circuit Board (PCB) fabricated using materials like FR4 or Rogers to support high-frequency operations. The PCB integrates the 8-element quad-band antenna array along with RF components such as Low Noise Amplifiers (LNA), Bandpass Filters (BPF), and power dividers/combiners for signal amplification, filtering, and distribution. Proper design ensures minimal losses and efficient signal transmission across all channels.

Implementation Points:

- **Hardware & Fabrication:** Antenna and RF circuits are implemented on PCB using suitable substrate materials.
- **Validation:** Performance is tested using Vector Network Analyzer (VNA), spectrum analyser, and RF signal generator.
- **Real-World Testing:** NavIC receiver module is used to evaluate system performance in practical conditions.

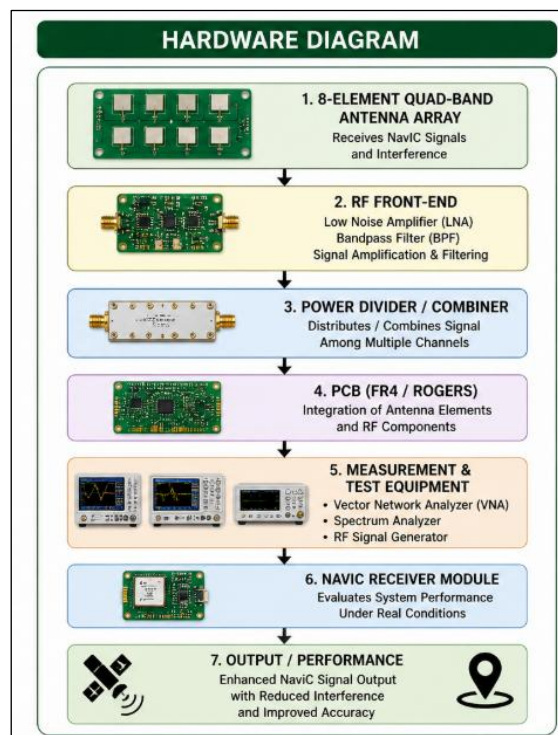


Fig 6.2: Hardware requirements.

CHAPTER 7: CIRCUIT DIAGRAM AND ITS EXPLANATION

The circuit design of the proposed system includes multiple antenna elements connected to individual RF chains. Each antenna element is followed by a low-noise amplifier to boost the weak received signals. Bandpass filters are used to isolate the desired frequency bands and prevent interference from other frequencies.

The amplified and filtered signals are then combined using power dividers and fed into subsequent processing stages. The circuit is designed to maintain proper impedance matching and minimize losses, ensuring efficient signal transmission.

The overall design enables independent processing of signals from each antenna element, which is essential for implementing spatial filtering techniques such as beamforming and null steering.

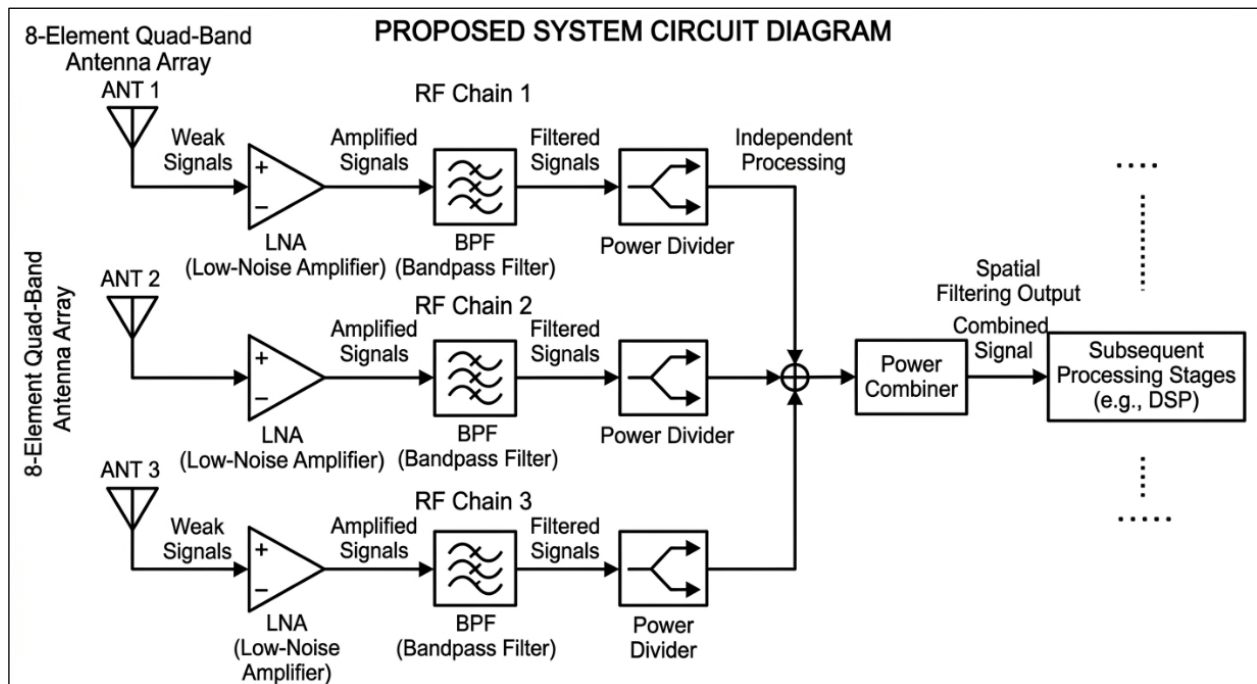


Fig 7.1: Circuit diagram for proposed system.

This circuit diagram illustrates the hardware signal flow designed to enhance NavIC satellite signals while suppressing environmental interference. The architecture is organized as follows:

1. 8-Element Quad-Band Antenna Array

- The system consists of 8 antenna elements (ANT1–ANT8) arranged in an array (linear/circular/planar).
- Each antenna receives:
 - Desired NavIC signals from satellites
 - Undesired signals (interference/jamming) from other sources
- Since signals arrive from different directions, each antenna receives them with slight phase and amplitude differences.
- This property is very important because it helps in Direction of Arrival (DoA) estimation and spatial filtering.
- Quad-band capability allows operation in L1, L2, L5, and S-band, improving reliability and redundancy.

2. Independent RF Chains (Parallel Architecture)

- Each antenna is connected to a separate RF chain, forming a multi-channel system (8 channels).
- All RF chains work in parallel, meaning signals are processed simultaneously.
- This parallel processing is essential for:
 - Comparing signals from different antennas
 - Applying beamforming and nulling techniques
- The architecture ensures no loss of spatial information.

3. Low Noise Amplifier (LNA)

- The first component in each RF chain is the LNA.
- NavIC signals are extremely weak (in the order of -130 dBm or lower), so amplification is necessary.

Functions:

- Amplifies weak signals to a usable level
- Maintains a low noise figure to preserve signal quality
- Improves Signal-to-Noise Ratio (SNR)

Output:

- Weak signals → Amplified signals

4. Bandpass Filter (BPF)

- After amplification, signals pass through a Bandpass Filter.

Functions:

- Selects only required frequency bands (NavIC bands)
- Rejects:
 - Out-of-band noise
 - Interference from other RF sources (WiFi, LTE, radar, etc.)

Importance:

- Prevents unwanted signals from entering further stages
- Protects ADC and DSP from overload

Output:

- Amplified signals → Filtered signals

5. Power Divider (Signal Splitting)

- Each filtered signal is fed into a Power Divider.

Functions:

- Splits the signal into multiple identical paths
- Allows:
 - Independent processing

- Signal comparison between channels

6. Independent Processing Paths

- After splitting, signals from each antenna are available for individual analysis.
- This stage maintains:
 - Phase information
 - Amplitude differences
- These parameters are later used for:
 - DoA estimation
 - Beamforming weight calculation

7. Power Combiner (Signal Combination)

- Signals from all RF chains are then combined using a Power Combiner.

Functions:

- Combines multiple antenna signals into a single output
- Performs spatial filtering based on signal alignment

Key Concept:

- If signals are in-phase (desired signals) → they add constructively
- If signals are out-of-phase (interference) → they cancel out

8. Spatial Filtering Output

- The combined signal is called the spatially filtered signal.
- Characteristics:
 - Strong desired NavIC signals
 - Reduced interference
 - Improved signal clarity

9. Subsequent Processing Stages (DSP)

- The combined signal is passed to Digital Signal Processing (DSP).

Operations performed:

- Direction of Arrival (DoA):
 - Calculates angle of incoming signals using phase differences
- Beamforming:
 - Focuses antenna gain in direction of satellites
- Null Steering:
 - Places nulls in direction of interference
- Weight Calculation:
 - Adjusts amplitude and phase of each antenna signal dynamically
- STAP (Advanced Processing):
 - Uses both spatial and time-domain filtering for better performance

10. Final Output

- Output is a clean, enhanced NavIC signal with:
 - High Signal-to-Noise Ratio
 - Reduced interference and jamming
 - Improved accuracy and reliability

CHAPTER 8: PROJECT IMPLEMENTATION WITH STEPS AND FLOW DIAGRAM

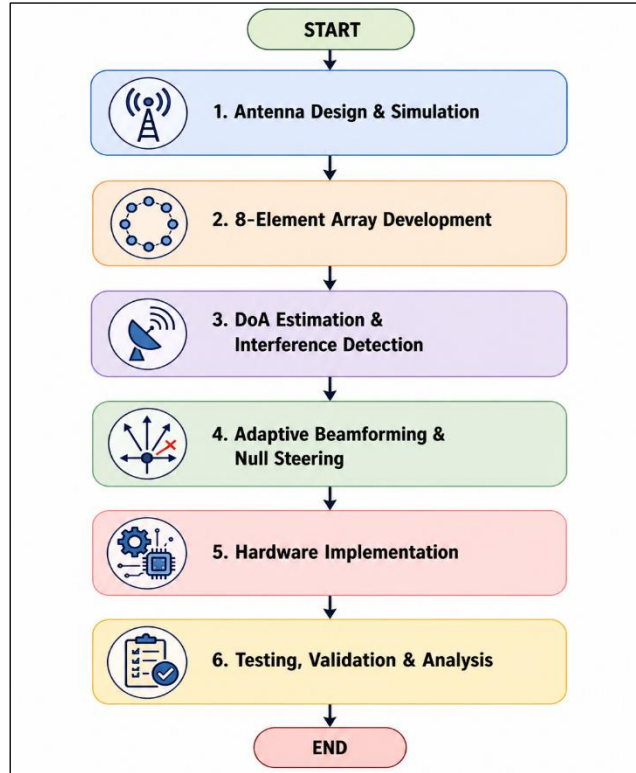


Fig 8.1: Flow chart for expected implementation steps.

1. Antenna Design & Simulation

In this initial stage, a compact quad-band antenna element is designed to operate efficiently across multiple NavIC frequency bands (L1, L2, L5, and S-band). Advanced electromagnetic simulation tools such as HFSS and CST Microwave Studio are used to model the antenna structure in a 3D environment. Various parameters like return loss (S11), gain, radiation pattern, bandwidth, and impedance matching are analyzed and optimized. The goal of this stage is to ensure that the antenna provides stable performance across all required bands with minimal losses and good efficiency before moving to physical implementation.

2. 8-Element Array Development

After designing a single antenna element, multiple identical elements are arranged to form an 8-element antenna array. The configuration of the array (such as spacing and geometry) is carefully selected to achieve higher gain and better directivity. This stage enhances the system's ability to receive signals more effectively by utilizing spatial diversity. Proper element spacing is maintained to avoid issues like mutual coupling, which can degrade performance. The array structure also enables spatial filtering, which is essential for distinguishing between desired signals and interference.

3. DoA Estimation & Interference Detection

In this stage, signal processing techniques are used to determine the Direction of Arrival (DoA) of incoming signals. By analysing phase and amplitude differences between signals received at different antenna elements, the system can identify the direction of both desired NavIC signals and interfering sources. Algorithms such as MUSIC or ESPRIT may be used for accurate estimation. This step is crucial because it provides the necessary information to differentiate between useful signals and jammers, enabling effective interference mitigation in later stages.

4. Adaptive Beamforming & Null Steering

Based on the DoA information, adaptive beamforming techniques are applied to focus the antenna array in the direction of the desired signal. At the same time, null steering is used to create nulls (points of minimum gain) in the direction of interference sources. This is achieved by adjusting the amplitude and phase (weights) of signals from each antenna element. As a result, the system enhances the desired signal while suppressing unwanted signals. This adaptive process continuously updates itself to handle dynamic environments where interference may change over time.

5. Hardware Implementation

In this stage, the designed system is physically implemented. The antenna array is fabricated on a Printed Circuit Board (PCB) using suitable materials such as FR4 or Rogers substrate. RF components like Low Noise Amplifiers (LNA), Bandpass Filters (BPF), power dividers, and connectors are integrated into the system. Proper layout design and impedance matching are ensured to minimize losses and maintain signal integrity. This stage converts the simulated design into a real working prototype.

6. Testing, Validation & Analysis

The final stage involves testing and evaluating the system to verify its performance. Instruments such as a Vector Network Analyzer (VNA) are used to measure antenna parameters like return loss and VSWR. A spectrum analyser and RF signal generator are used to analyse signal quality and interference handling. The system is also tested using a NavIC receiver module to evaluate real-time performance. The results are analysed to confirm that the system effectively reduces interference and improves signal quality. Any required improvements are identified and implemented based on the analysis.

CHAPTER 9: RESULTS AND DISCUSSION

The proposed anti-jamming antenna system is expected to achieve good performance across all targeted NavIC frequency bands (L1, L2, L5, and S-band). The return loss (S11) is expected to be below -10 dB, indicating proper impedance matching and minimal signal reflection. The Voltage Standing Wave Ratio (VSWR) is expected to be less than 2, ensuring efficient power transfer and stable antenna operation.

The use of an 8-element antenna array is expected to provide higher gain and improved directivity compared to a single antenna element. This enhances the system's ability to receive weak satellite signals effectively. The antenna array is also expected to produce stable radiation patterns and enable spatial filtering, which helps in distinguishing between desired signals and interference.

With the implementation of adaptive beamforming and null steering techniques, the system is expected to suppress interference effectively. It can form beams in the direction of desired signals while creating deep nulls in the direction of jammers. This improves the Signal-to-Noise Ratio (SNR) and ensures more accurate and reliable navigation performance even in challenging environments.

The results will also highlight the advantages of quad-band operation, which reduces the need for multiple antennas and simplifies the system design. However, challenges such as mutual coupling between antenna elements and maintaining consistent performance across multiple frequency bands may arise, which need to be addressed through proper design and optimization.

The preliminary antenna simulations using ANSYS HFSS provided valuable insights into the design and performance of the proposed system. The 1×1 CSMA antenna showed good impedance matching with return loss below -10 dB and VSWR less than 2 at around 11.4 GHz, along with a gain of approximately 7.5 dB. The 1×2 microstrip patch antenna array demonstrated dual-band operation at 2.4 GHz and 4.0 GHz with improved gain and directivity. These results helped in validating key antenna parameters and served as a foundation for the design and optimization of the final 8-element quad-band anti-jamming antenna array.

CHAPTER 10: CONCLUSION

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CHAPTER 11: FUTURE SCOPE

The future scope of this project focuses on improving the overall performance, capacity, and adaptability of the anti-jamming antenna system to meet the demands of evolving RF environments. One important direction is increasing the number of antenna elements beyond the current 8-element array. A larger array can provide better spatial resolution, sharper beamforming, and improved interference suppression capability. This will enhance the system's ability to detect and reject multiple jamming sources simultaneously, making it more effective in complex and high-interference scenarios.

Another significant area of development is the transition from simulation-based implementation to real-time hardware execution. Implementing the system on platforms such as Field Programmable Gate Arrays (FPGA) or Software Defined Radio (SDR) can greatly improve processing speed and reduce latency. Real-time processing allows the system to respond instantly to dynamic interference conditions, making it more suitable for practical applications such as defence, aviation, and autonomous systems. This shift will also enable on-field testing and deployment in real-world environments.

The integration of Artificial Intelligence (AI) and Machine Learning (ML) presents another promising advancement for this system. By incorporating intelligent algorithms, the system can automatically detect, classify, and adapt to different types of interference such as jamming and spoofing. Machine learning models can be trained to recognize signal patterns and optimize beamforming strategies in real time. This will make the system more autonomous, efficient, and capable of handling complex and unpredictable signal environments without manual intervention.

Finally, future work can focus on the miniaturization and optimization of hardware components to make the system more compact and portable. Reducing the size of the quad-band antenna elements and RF front-end circuitry will allow the system to be integrated into smaller platforms such as handheld devices, drones, and tactical UAVs. This will expand its usability across various applications while maintaining high performance. Overall, these advancements will make the anti-jamming system more powerful, flexible, and suitable for next-generation navigation technologies.

CHAPTER 12: APPLICATIONS

The proposed anti-jamming antenna system has critical applications in **defence and GNSS-based systems**, where secure and reliable navigation is essential. In military operations, navigation systems are often targeted by electronic warfare techniques such as jamming and spoofing. The proposed system provides a strong layer of protection by ensuring accurate Position, Navigation, and Timing (PNT) data even in hostile environments. This improves mission reliability, enhances situational awareness, and supports secure communication and coordination of defence assets.

In the **UAV and drone industry**, the system plays an important role in maintaining signal integrity and navigation accuracy. Drones are widely used for applications such as surveillance, logistics, mapping, and disaster monitoring. However, they often operate in environments with high RF interference or intentional jamming. The use of an 8-element antenna array enables reliable navigation by suppressing interference and maintaining strong signal reception. This ensures safe operation of drones in urban areas, remote locations, and critical missions.

In the **aviation and maritime sectors**, precise and uninterrupted navigation is extremely important for safety. The proposed system enhances the reliability of aircraft navigation systems, especially during critical operations such as landing and takeoff, where accurate positioning is required. Similarly, in maritime navigation, it helps ships maintain correct routes and avoid collisions, even in the presence of signal interference. By preventing signal loss and improving accuracy, the system reduces safety risks and ensures smooth operation in these sectors.

The system also has significant importance in **national infrastructure and satellite communication systems**, such as ISRO ground stations and GNSS-based networks. It ensures continuous and accurate delivery of PNT services, which are essential for applications like synchronized power grids, telecommunications, transportation systems, and disaster management. Reliable navigation and timing information support the smooth functioning of these systems, making the proposed anti-jamming antenna a key technology for modern infrastructure and large-scale operations.

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APPENDIX

Implementation of 1×1 CSMA antenna

The 1×1 CSMA antenna was designed and simulated using ANSYS HFSS to evaluate its performance. The antenna operates at approximately 11.4 GHz with return loss below -10 dB, indicating good impedance matching. The VSWR value is less than 2 at this frequency, confirming minimal power reflection. The design consists of a single patch element on a dielectric substrate with a ground plane, and the radiation pattern shows directional behavior with a gain of about 7.5 dB. Overall, the antenna demonstrates efficient single-band operation suitable for wireless applications.

1. Antenna Design (HFSS Model)

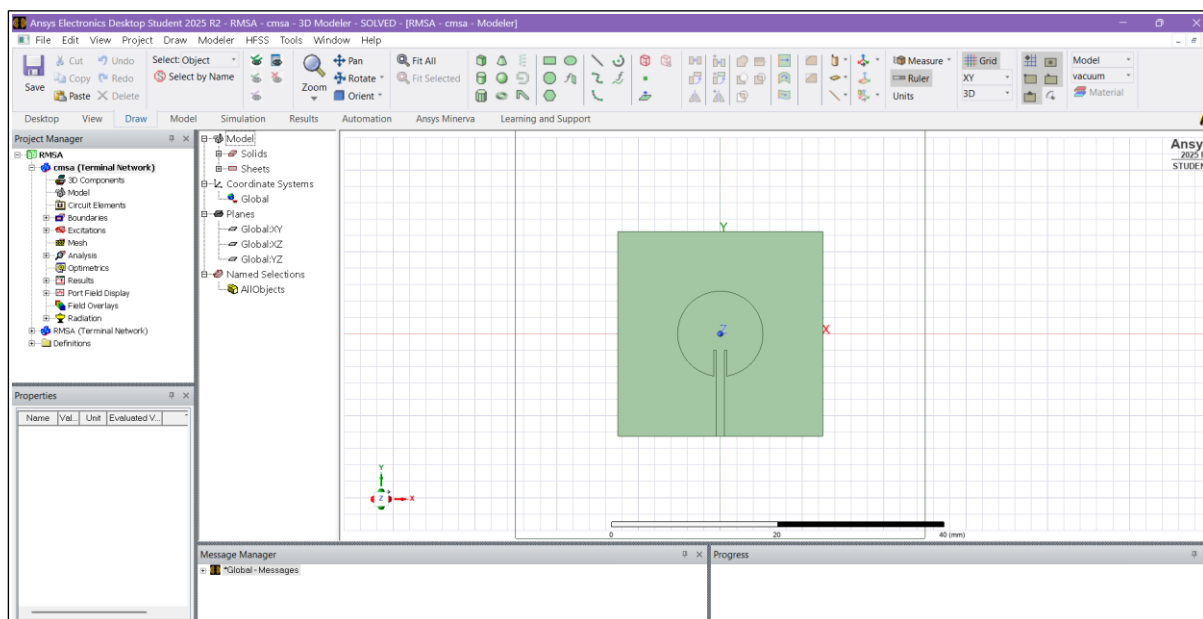


Fig (a): 1×1 Antenna array design.

- Designed using ANSYS HFSS
- Structure includes:
 - Single patch element (1×1)
 - Dielectric substrate
 - Ground plane
 - Air box (simulation boundary)

- Represents basic antenna structure
- Used as a reference for array design

2. S-Parameter (Return Loss) Plot

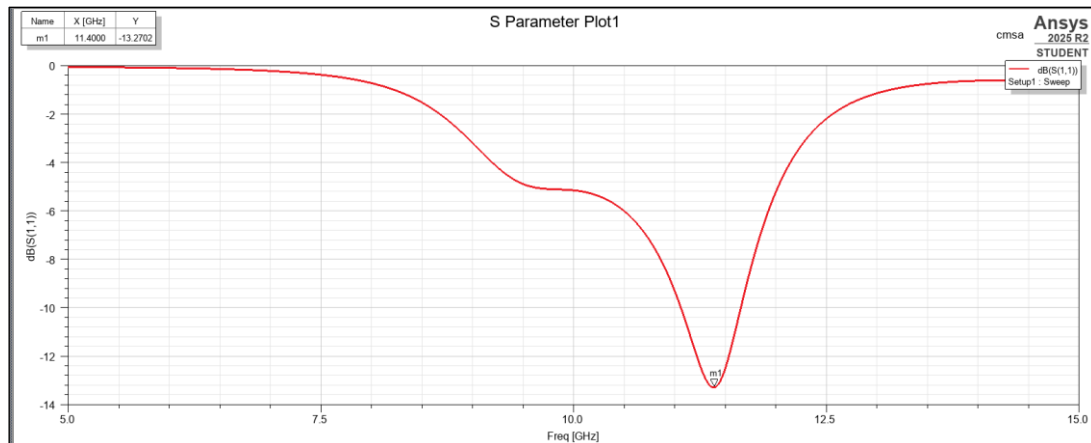


Fig (b): Return loss plot.

- Shows reflection coefficient (S11) vs frequency
- Dip indicates resonant frequency
- Resonance observed at:
~11.4 GHz
- Return loss is below -10 dB → good matching
- Less reflected power → better radiation efficiency
- Indicates single-band operation

3. VSWR Plot

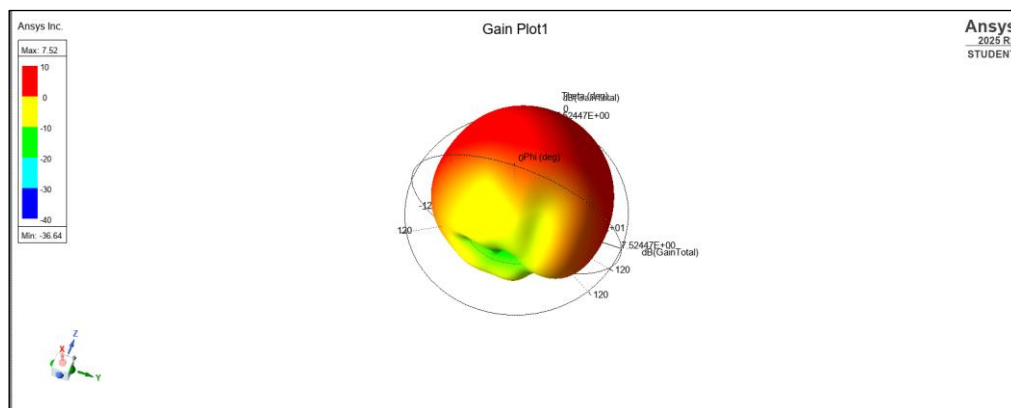


Fig (c): VSWR plot.

- Shows Voltage Standing Wave Ratio vs frequency
- Ideal VSWR = 1
- At ~11.4 GHz:

VSWR $\approx$ 1–2 $\rightarrow$ acceptable performance

- VSWR < 2 confirms low reflection and efficient power transfer
- Supports S11 results

4. Gain Plot (Radiation Pattern)

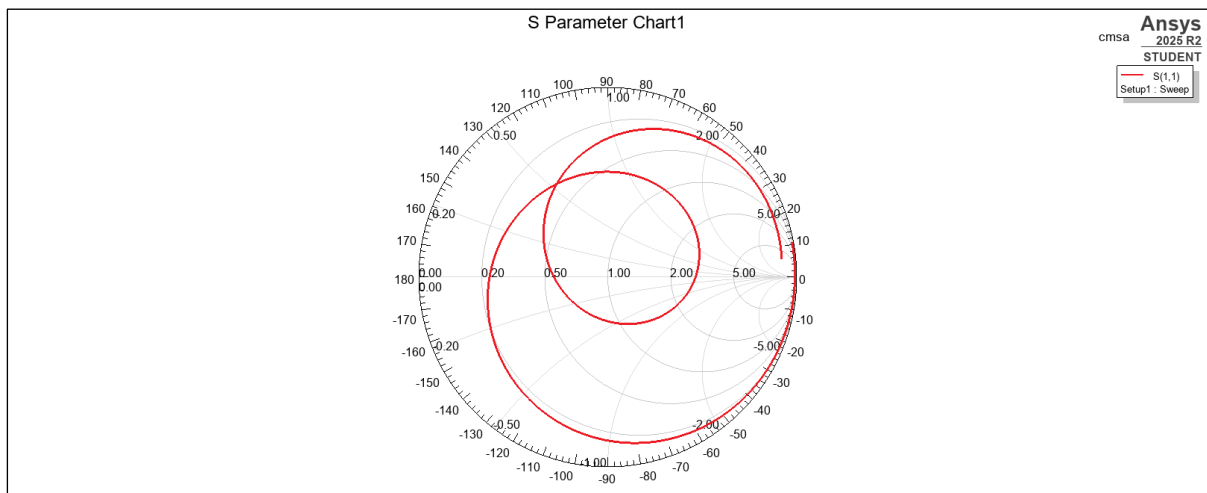


Fig (d): Gain plot.

- Shows 3D radiation pattern of antenna
- Maximum gain $\approx$ 7.5 dB
- Indicates directional radiation
- Main lobe = direction of maximum radiation
- Suitable for wireless communication

Implementation of 1×2 microstrip patch antenna array

The 1×2 microstrip patch antenna array was designed and simulated using ANSYS HFSS to evaluate its performance. The antenna operates at approximately 2.4 GHz and 4.0 GHz with return loss below -10 dB, indicating good impedance matching. The VSWR values are less than 2 at these frequencies, confirming minimal power reflection. The design consists of two

patch elements on a dielectric substrate with a ground plane, and the radiation pattern shows directional behavior with a gain of about 5.7 dB. Overall, the antenna demonstrates efficient dual-band operation suitable for wireless applications.

1. Antenna Design (HFSS Model)

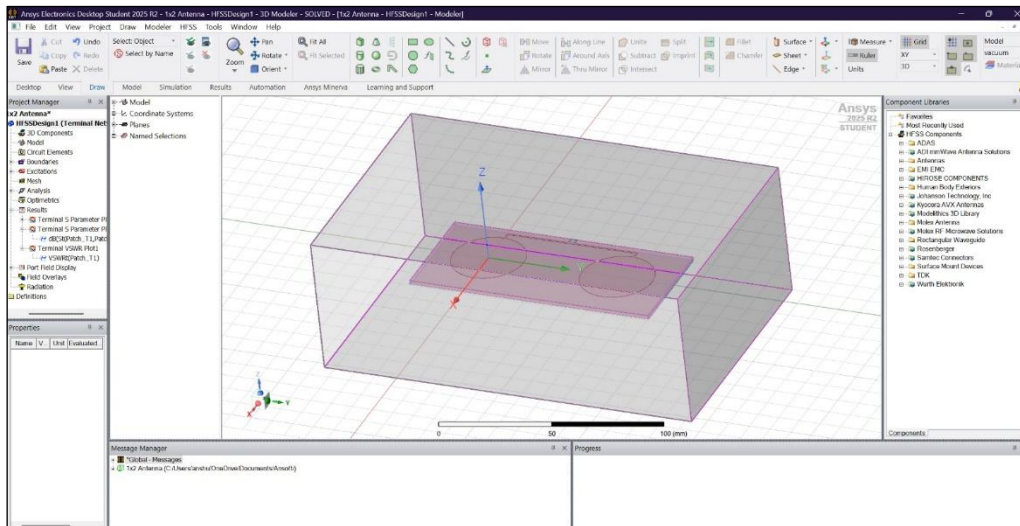


Fig (e): 1x2 Antenna array design.

- Designed using ANSYS HFSS
- Structure includes:
 - Two patch elements (1×2 array)
 - Dielectric substrate
 - Ground plane
 - Air box (simulation boundary)
- Array configuration improves:
 - Gain
 - Directivity
- Represents physical layout of antenna

2. S-Parameter (Return Loss) Plot

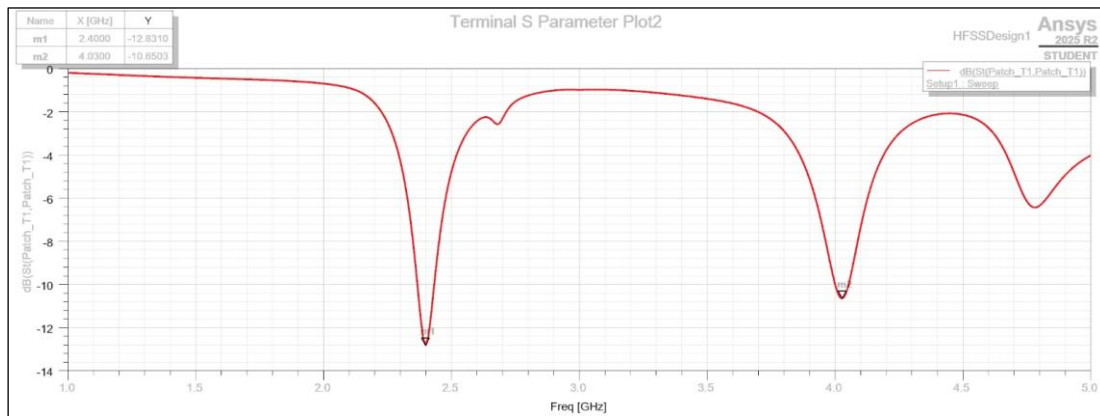


Fig (f): Return loss plot.

- Shows reflection coefficient (S11) vs frequency
- Dips indicate resonant frequencies
- Resonance observed at:
 - ~2.4 GHz
 - ~4.0 GHz
- Return loss is below -10 dB at these points → good matching
- Less reflected power → better radiation efficiency
- Indicates dual-band operation

3. VSWR Plot

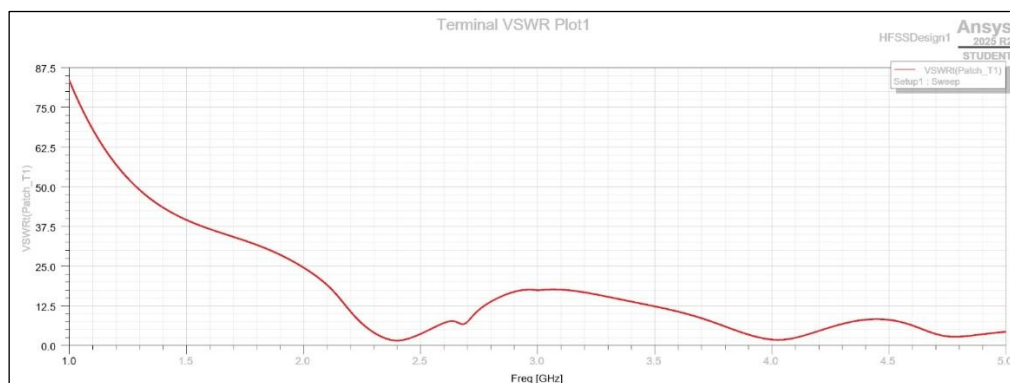


Fig (g): VSWR plot.

- Shows Voltage Standing Wave Ratio vs frequency
- Ideal VSWR = 1 (perfect matching)
- At ~2.4 GHz and ~4 GHz:
 - VSWR $\approx$ 1–2 $\rightarrow$ acceptable performance
- VSWR < 2 confirms low reflection and efficient power transfer
- Supports S11 results

4. Gain Plot (Radiation Pattern)

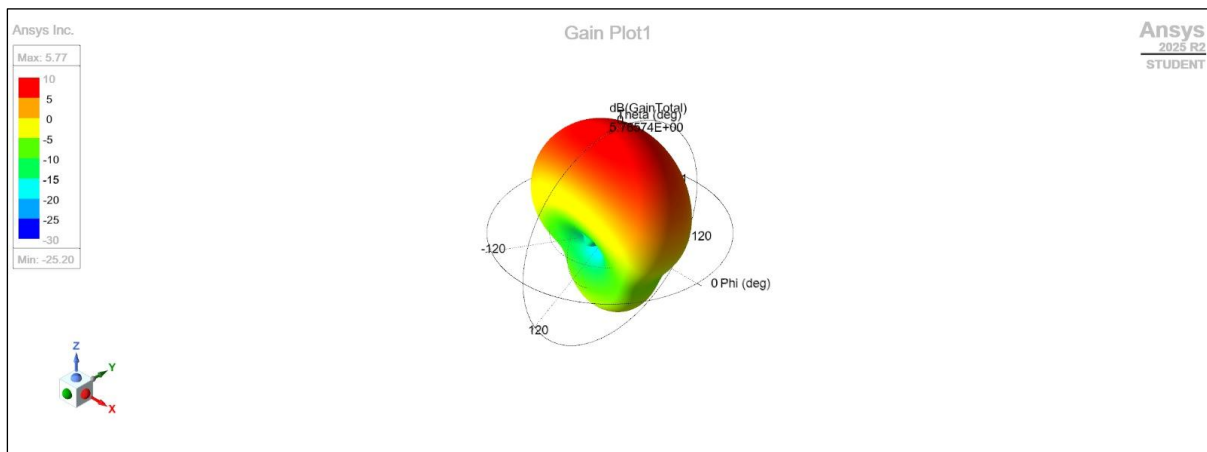


Fig (h): Gain plot.

- Shows 3D radiation pattern of antenna
- Maximum gain $\approx$ 5.7 dB
- Indicates directional radiation
- Main lobe = direction of maximum radiation
- Suitable for targeted wireless communication

The 1×1 CSMA antenna was first designed as a preliminary step to analyse and validate key antenna parameters, which were later extended and optimized for array antenna design.

The 1×2 antenna array was first designed as a preliminary step to analyse and validate key antenna parameters, which were later extended and optimized for the final 8-element quad-band anti-jamming antenna design.

PLAN OF ACTION FOR THE NEXT SEMESTER

In the next semester, the primary focus of the project will be on the practical implementation and validation of the proposed 8-element quad-band anti-jamming antenna system. After completing the design and simulation phases in the current semester, the work will progress towards translating the simulated model into a functional hardware prototype.

The initial phase will involve the fabrication of the quad-band antenna elements on a suitable substrate such as FR4 or Rogers material. Careful attention will be given to maintaining dimensional accuracy to ensure that the fabricated antenna operates at the desired L1, L2, L5, and S-band frequencies. Following this, the 8-element antenna array will be constructed, ensuring proper element spacing and alignment to minimize mutual coupling and achieve the desired radiation characteristics.

Subsequently, the RF front-end circuitry will be developed, which includes integration of components such as low-noise amplifiers (LNA), bandpass filters, and power dividers. These components will be assembled and connected to the antenna array using appropriate RF connectors and cables. The goal of this stage is to ensure efficient signal reception and conditioning before further processing.

In parallel, the signal processing algorithms developed in MATLAB or Python, such as Direction of Arrival (DoA) estimation and adaptive beamforming techniques, will be prepared for real-time or near real-time implementation. If feasible, these algorithms may be tested using platforms such as Software Defined Radio (SDR) or microcontroller-based systems.

Once the hardware setup is completed, the system will undergo testing and validation using laboratory instruments such as a Vector Network Analyzer (VNA) and spectrum analyzer. Key performance parameters including return loss (S_{11}), VSWR, gain, radiation pattern, and interference suppression capability will be measured and analyzed.

Finally, the results obtained from hardware testing will be compared with simulation results to evaluate system performance. Necessary modifications and optimizations will be carried out to improve accuracy and efficiency. The semester will conclude with documentation, result analysis, and final project demonstration, ensuring that the system meets the intended objectives of robust anti-jamming and reliable NavIC signal reception.

PROJECT BUDGET

Category	Details	Estimated Cost (INR)
Antenna Fabrication	8-element quad-band PCB array, connectors, mechanical supports, shielding, RF substrates, RF cables.	50,000=00
Simulation & Design Software	CST/HFSS/ADS educational or institutional licenses, EM simulation, beamforming simulations.	5,00,000=00
Microcontroller / SDR / Beamforming Hardware	STM32 / Arduino / USRP B200/B210 / HackRF + FPGA/DSP boards for beamforming.	20,000=00
Testing & Measurement Instruments	VNA (mid-range lab-grade) + Spectrum Analyzer (bench-top 1–3 GHz) + Signal Generator.	10,00,000=00
Jammer / NavIC CDMA Signal Emulator	Lab-grade low-power interference generator, CDMA/NavIC test signal setup.	50,000=00
Field Testing / Integration	Mini test setup for validating anti-jamming performance, NavIC receiver integration.	25,000=00
Miscellaneous / Consumables	Cables, adapters, soldering tools, mechanical parts, protective enclosure.	10,000=00
Contingency (≈10%)	Unforeseen costs, shipping, small additions.	25,000=00
Others	Stationery & Documentation Printing, plotting diagrams, report preparation materials. Lab notebooks, markers, labels	10,000=00

Total Estimated Budget: ₹15–20 Lakhs